

GRAACE-SIM: A user friendly PGAANAA simulation framework

Modular | Configurable | Reproducible

User Inputs

Experiment Definition

- Source parameters
- Sample properties & geometry
- Detector configuration
- Shielding & surroundings
- Run settings

Predefined Libraries

- Sources
- Detectors
- Materials
- Geometries
- Shielding

User-Defined Inputs

- Custom geometries
- Materials
- Detectors
- Run settings

Python Control Layer

Configuration - Validation - Execution Management - Post-Processing

Collect & Manage Outputs

Manage Runs (Batch / HPC)

Build Geometry & Components

Parse & Validate Inputs

GEANT4 Simulation Core

Particle Transport & Detector Response

Detector Response (HPGe, etc.)

Particle Transport (EM + Hadronic)

Gamma Production (Activation)

Neutron Interactions

Outputs

Visualization

- Counts vs. energy (keV)
- Multi-run spectrum overlays
- Self-shielding ratio plots
- Detector geometry pictures
- Geometry with particle tracks

Simulation Results

- Gamma-ray spectra
- Pulse height spectra
- Neutron flux maps
- Detector responses
- Run metadata

Analysis & Metrics

- Sensitivity
- Minimum flux estimates
- Background estimates
- Uncertainty